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LANGUAGE



In this chapter, we step into the messy world of human language. We'll start by discussing the differences between language and math that make language algorithms difficult. We'll continue by building a space insertion algorithm that can take any text in any language and insert spaces wherever they're missing. After that, we'll build a phrase completion algorithm that can imitate the style of a writer and find the most fitting next word in a phrase.

The algorithms in this chapter rely heavily on two tools that we haven't used before: list comprehensions and corpuses. *List comprehensions* enable us to quickly generate lists using the logic of loops and iterations. They're optimized to run very quickly in Python and they're easy to write concisely, but they can be hard to read and their syntax takes some getting used to. A *corpus* is a body of text that will “teach” our algorithm the language and style we want it to use.